



Interim Presentation

Development and Validation of a Dexterous Hand System

··· Dexhand Designed by The Robot Studio···

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Overall Workflow

Mechanical Assembly

Electronics & Embedded Testing

Simulation

Project Background

- DexHand is used in this project as an open-source dexterous hand platform
- The project combines simulation, hardware implementation, and intelligent control exploration

Project Goal

- Reproduce the main workflow of DexHand as a course project
- Understand its mechanical, electronic, and control architecture
- Prepare for later integrated testing and intelligent control development

Main Work Packages

- Mechanical Assembly
- Electronics and Embedded Testing
- Simulation / RL Exploration

Interim Progress Focus

- ROS2 simulation workflow reproduced
- Hardware architecture and actuator selection analyzed
- Mechanical assembly and PCB preparation started



Overall Progress of Mechanical structure

Finger

↓ 100%

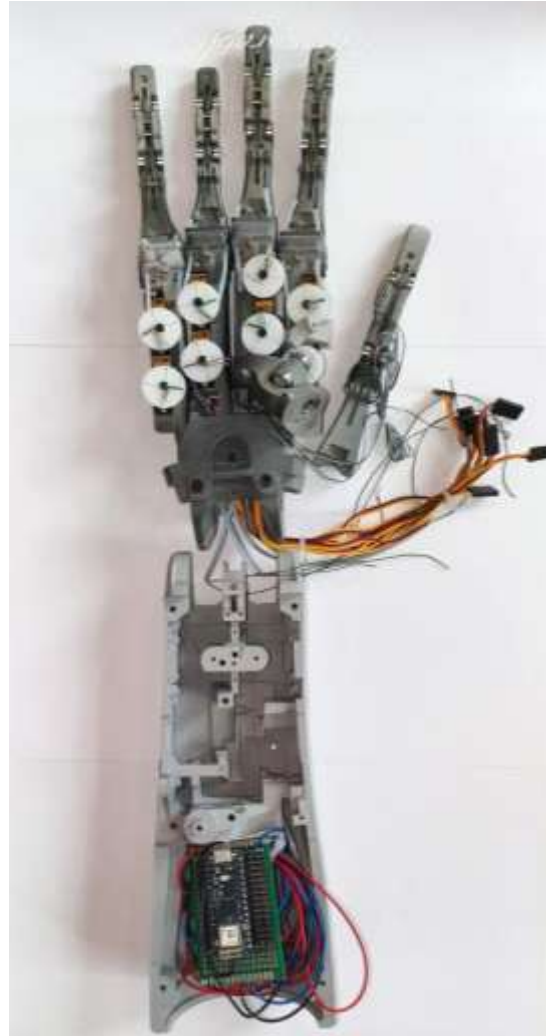
Palm

↓ 90%

Waist

↓ 20%

Forearm



75%

Ideal Effect

Standard digit size is 100. There are three finger sizes used in this hand: 90, 100 & 105

Fore Finger 100

Middle Finger 105

Ring Finger 100

Little Finger 90

All fingers and the thumb consist of three finger sections: the tip, MLP and PLP.

TIP

MLP – Middle Phalanges

PLP – Proximal Phalanges

Thumb 100

CMC – Carpometacarpal - the thumb knuckle joint consists of two pieces.

The fingers connect to the palm sections via the Knuckles.

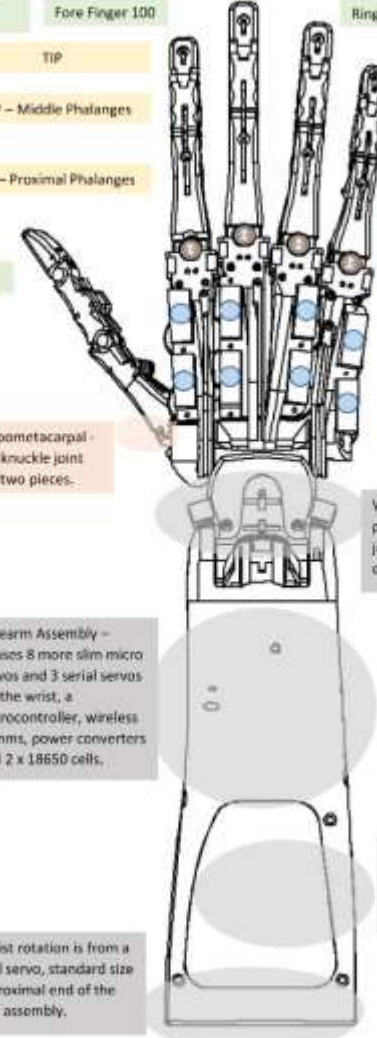
8 x eMax ES33xx micro slim servos drive the fingers at the knuckle joint via tendons.

Wrist Assembly – has two pitch joints and a yaw joint to give the full range of human movement.

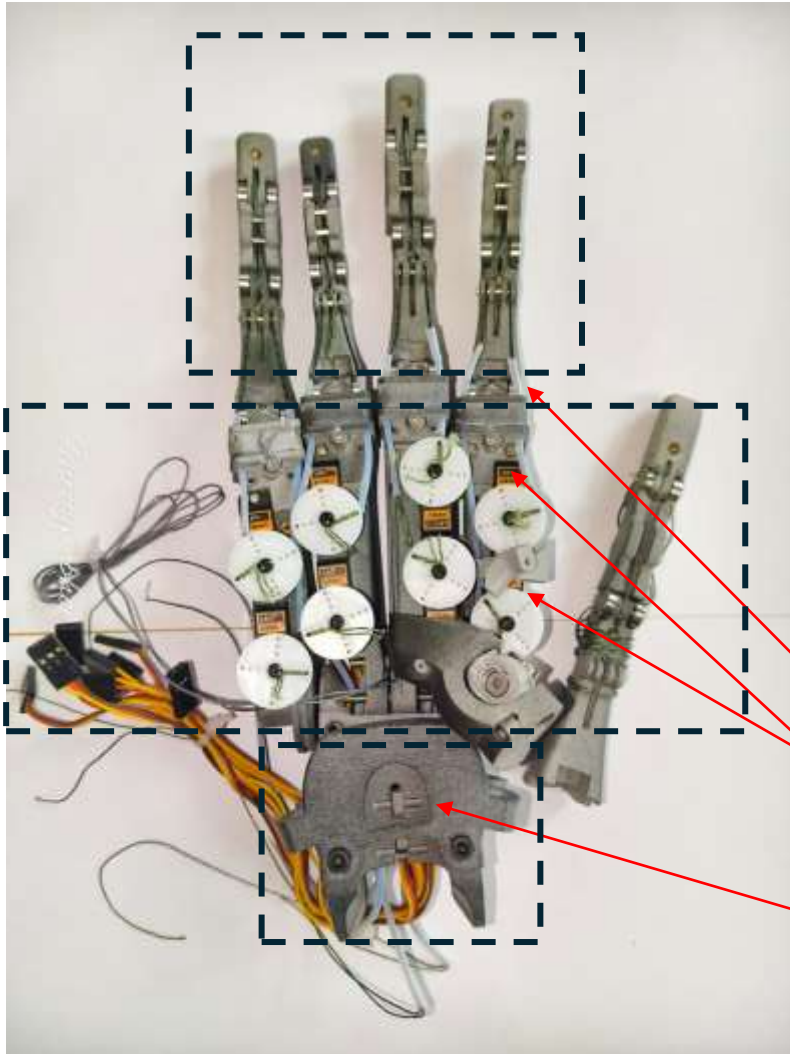
Forearm Assembly – houses 8 more slim micro servos and 3 serial servos for the wrist, a microcontroller, wireless comms, power converters and 2 x 18650 cells.

Axial wrist rotation is from a 3" serial servo, standard size at the proximal end of the forearm assembly.

Battery Lid – removes for access to 2 x Li-Ion 18650 cells.



Finger & Palms & Wrist



Finger

Distal interphalangeal (DIP) joint flexion/extension → **1 DOF**

Proximal interphalangeal (PIP) joint flexion/extension → **1 DOF**

Metacarpophalangeal (MCP) joint flexion/extension and abduction/adduction → **2 DOF**

5 fingers × 4 DOF per finger = 20 DOF



Palm

The palm houses key components such as springs and servo motors.

Springs provide the restoring force for finger motion.

Tubes protect the tendons and prevent cable entanglement.

Each finger is actuated by two servo motors for lateral motion.

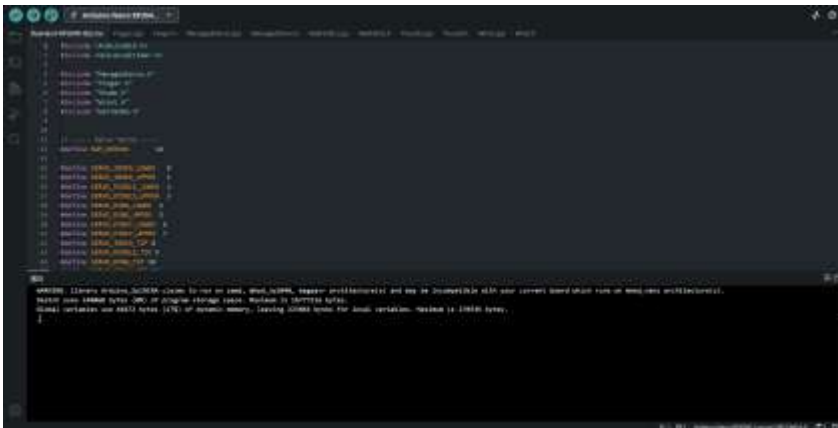
Wrist

The wrist supports **pitch** and **yaw**, corresponding to **1 + 1 = 2 DOF**.

Main Control



Assembled control board



Build and upload

Completed Work

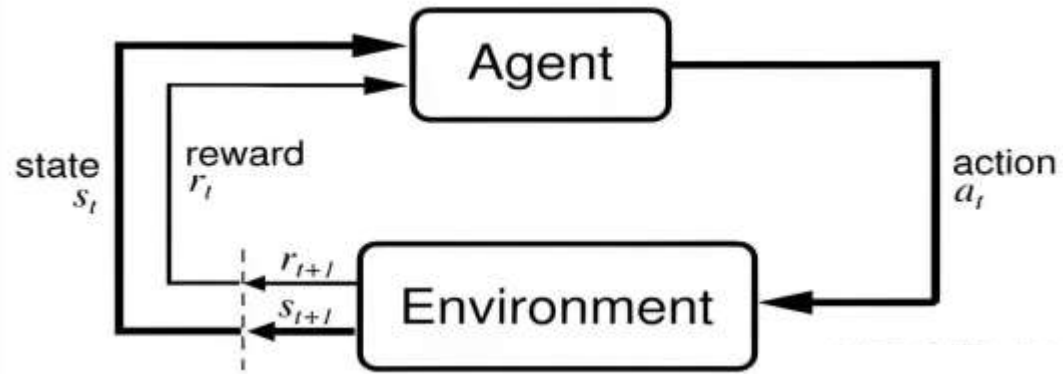
- Completed PCB soldering and control board preparation
- Finished firmware flashing for the controller board
- Expanded the related circuit connections for hardware testing

Current Status

- The control board can be programmed successfully
- Basic circuit expansion has been completed
- The hardware platform is ready for actuator-level testing

Next Step

- Connect the board to the servo system
- Verify signal output and actuator response
- Continue integrated debugging with the mechanical structure



Basic RL Workflow

RL Learning

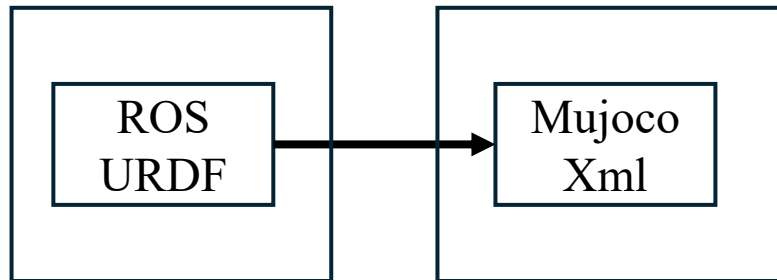
- Studying the basic concepts of reinforcement learning
- Learning the formulation of state, action, and reward
- Reviewing RL workflows for dexterous hand control

Environment Setup

- Started building the simulation environment in MuJoCo
- Exported the dexterous hand model from ROS **URDF** to **XML**

Next Step

- Validate the imported hand model in MuJoCo
- Refine the simulation environment for control tasks
- Begin preliminary RL-related experiments on hand motion



Model Conversion Pipeline

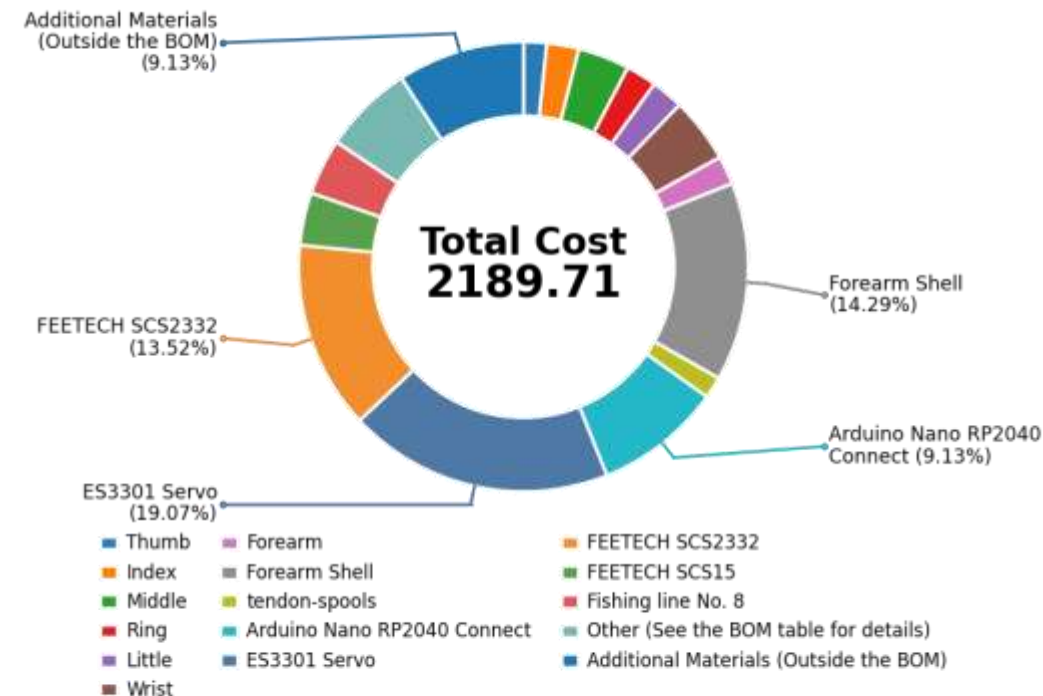
Work Area	Planned Tasks
Mechanical	Complete the full hand assembly; finish tendon routing and joint adjustment
Electronics	Continue control board and actuator testing; integrate the servo system with the hardware platform
RL / Intelligent Control	Validate the MuJoCo model; refine the environment for later intelligent control exploration
Outcome Area	Target Result
Hand Prototype	A functional dexterous hand prototype with integrated mechanical and electronic modules
System Verification	Verified actuator control and basic hand motion capability
Simulation-to-Hardware	Initial validation of the DexHand workflow from simulation to physical implementation





Components		Cost	Ratio	Satus
Finger	Thumb	36.41	1.66%	Purchased
	Index	50.68	2.31%	Purchased
	Middle	81.34	3.71%	Purchased
	Ring	48.41	2.21%	Purchased
	Little	50.77	2.32%	Purchased
Wrist		102.67	4.67%	Purchased
Forearm		45.05	2.06%	Purchased
Forearm Shell		313	14.29%	Purchased
tendon-spools		34	1.55%	Purchased
Arduino Nano RP2040 Connect		1*200=200	9.13%	Purchased
ES3301 Servo		16*26.1=417.6	19.07%	Purchased
FEETECH SCS2332		2*148=296	13.52%	Purchased
FEETECH SCS15		1*83=83	3.79%	Purchased
Fishing line No. 8		1*88.4=88.4	4.04%	Purchased
Other (See the BOM table for details)		142.38	6.50%	Purchased
Additional Materials (Outside the BOM)		About 200	9.13%	In Preparation
Total		1989.71+200	100%	

Other: 4cm*6cm Arduino通用原型板/2.54 排母20pin/2.54 排针20pin/圆柱销套件/M2 螺丝及垫片/螺纹嵌件
 m2*4*3.5/12编100m灰色/6103滚珠轴承/263滚珠轴承/深沟球轴承6702ZZ/693zz 384/m3*60/m3*70





Thank you for listening!

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